

M11 / ASSESSMENT

M11 Module Revision Test

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

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Save the full attempt before opening protected answers. This samples all ten lessons; there is no formal Gate after M11.

1. DE11-01 - Row vs columnar storage; where Parquet fits

For three workloads - point lookup of one customer, nightly sum of amount by region across millions of rows, and a vendor handoff to a nontechnical partner - choose row-oriented database/CSV/Parquet and defend the trade-off. Then state what Parquet buys the nightly analytical case.

2. DE11-02 - Parquet schema, row groups and metadata

Using `parquet_metadata_fixture.json`, decide which row groups could be skipped for (a) `order_date >= 2026-09-03`, (b) `amount > 195`, and (c) `region = 'East'` when the fixture has no region min/max. Explain exactly what evidence supports each decision.

3. DE11-03 - Projection, predicates and analytical reads

For these three queries, list output columns, filter columns, candidate row groups from `parquet_metadata_fixture.json`, and execution evidence for actual pruning: A) return `order_id, amount` where `order_date >= 2026-09-03`; B) return `order_id` where `amount > 195`; C) return `order_id, amount` where `region = East`. Do not use runtime alone as the only proof.

4. DE11-04 - Compression, small files and compaction

Given 40 files of 3 MB and 4 files of 600 MB, propose separate actions for the two groups and name one risk of compacting everything into a single huge file.

5. DE11-05 - Partitioning strategy and over-partitioning

Choose among no partitioning, date, region, customer_id and order_id for a hypothetical 2 TB order table. State what workload facts you would demand before deciding. Your answer must include a cardinality/file-size check, not just a query-filter example.

6. DE11-06 - Lake vs warehouse vs lakehouse

For a small company with raw event archives, curated finance tables and a BI workload, sketch lake, warehouse and lakehouse options. For each, state where raw bytes live, how table correctness is enforced and how analysts query serving data.

7. DE11-07 - Delta tables, transaction log and ACID concepts

Draw version 0 and version 1 as sets of active files plus log commits. Explain atomicity in one sentence and state what a reader should observe if a writer fails before committing version 2.

8. DE11-08 - MERGE/upsert and current-state records

Before writing MERGE SQL, specify target grain, match key, source deduplication/order rule, matched action, not-matched action, and three post-merge controls. Then compare target row count, unique keys and total amount to the expected fixture controls.

9. DE11-09 - Schema enforcement, controlled evolution and compatibility

Classify five changes - add nullable source_system, rename customer_id, int to double, string to struct, drop region - into 'likely additive/controlled', 'needs consumer migration', or 'incompatible/full rewrite review'. Defend each with downstream impact, not just engine capability.

10. DE11-10 - Time travel, retention, VACUUM safety and medallion checkpoint

Write a maintenance runbook that includes history inspection, time-travel requirement, retention owner, VACUUM DRY RUN, safety check, and post-maintenance validation. Then sketch Bronze/Silver/Gold contracts for the M11 orders: what is preserved, what is validated, and what is served.